University of Regina
Winter 2026

VISUAL EDUCATION SYSTEM

INTERACTIVE MEMORY-PALACE PROTOTYPE

CTCH 312 CONCEPT & STORY MILESTONE

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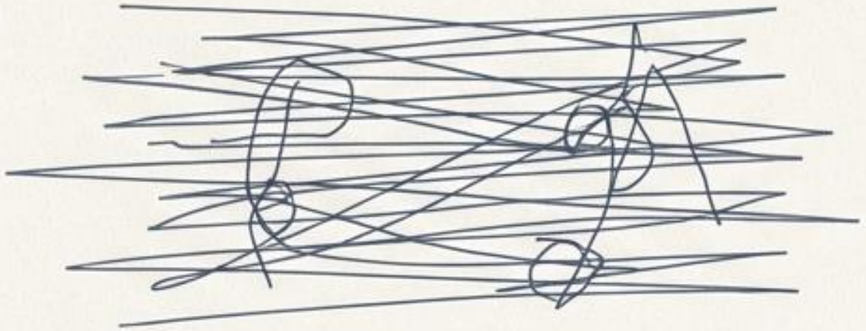
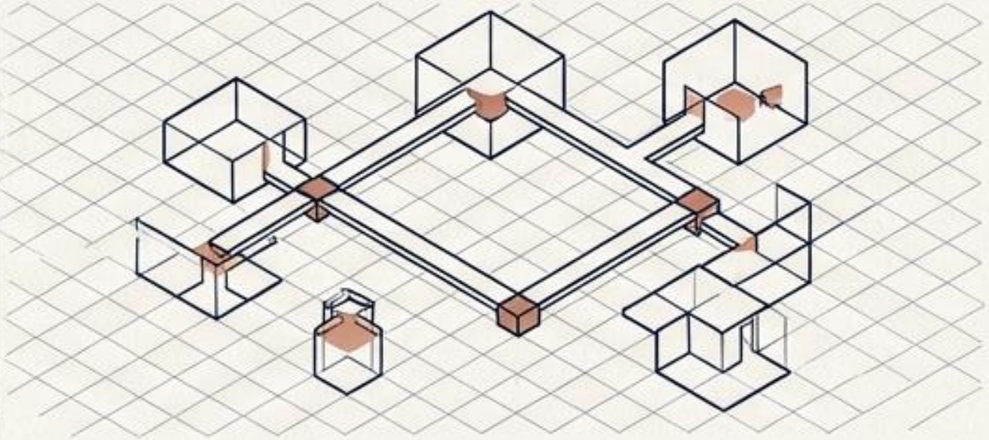
The Challenge of Retention

Learners often struggle to organize abstract information in memorable ways.

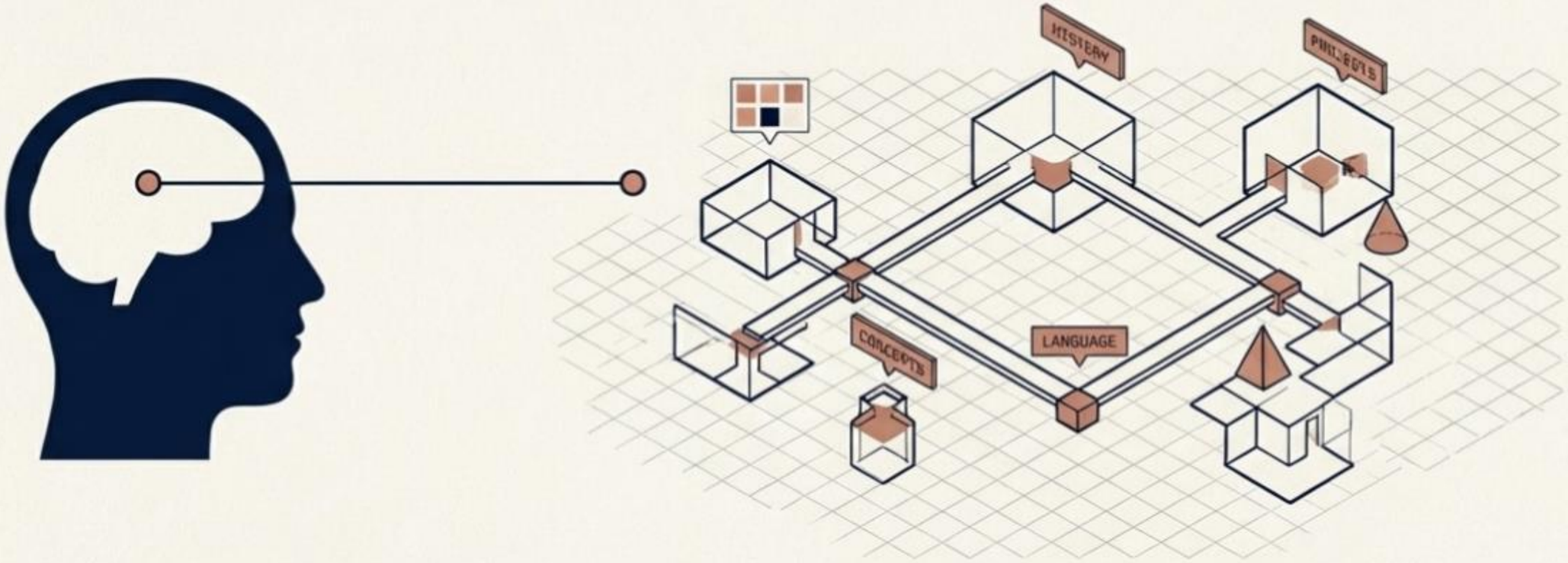




Standard Note-Taking	Digital Memory Palace
Linear & Abstract	Spatial & Visual
Fleeting recall	Durable retention

Project Hypothesis: Can a customizable digital palace improve learning retention?

A Personalized Memory Structure

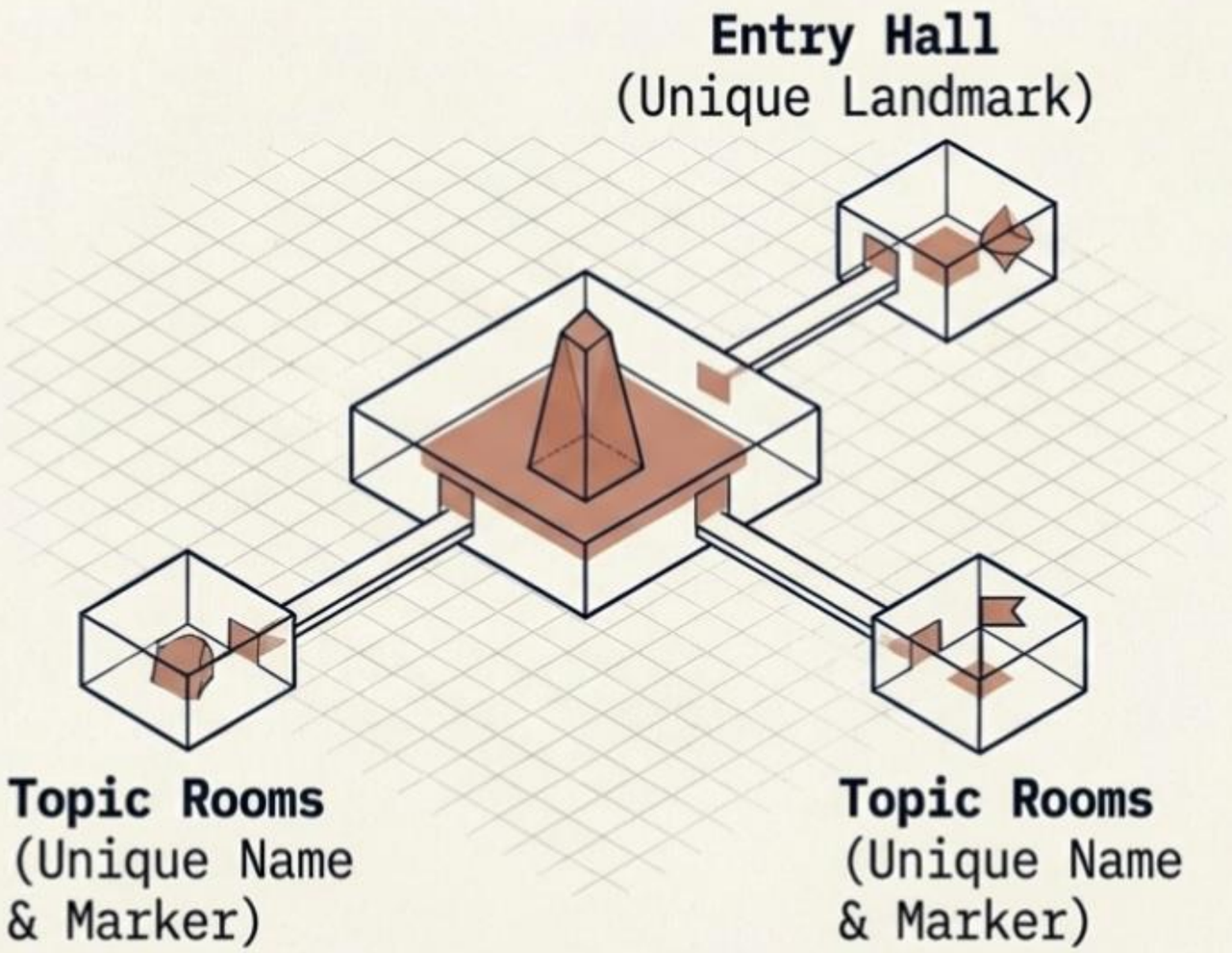


Enter: The user navigates a personalized, digital palace.

Categorize: Each room represents a memory category, topic, or concept cluster.

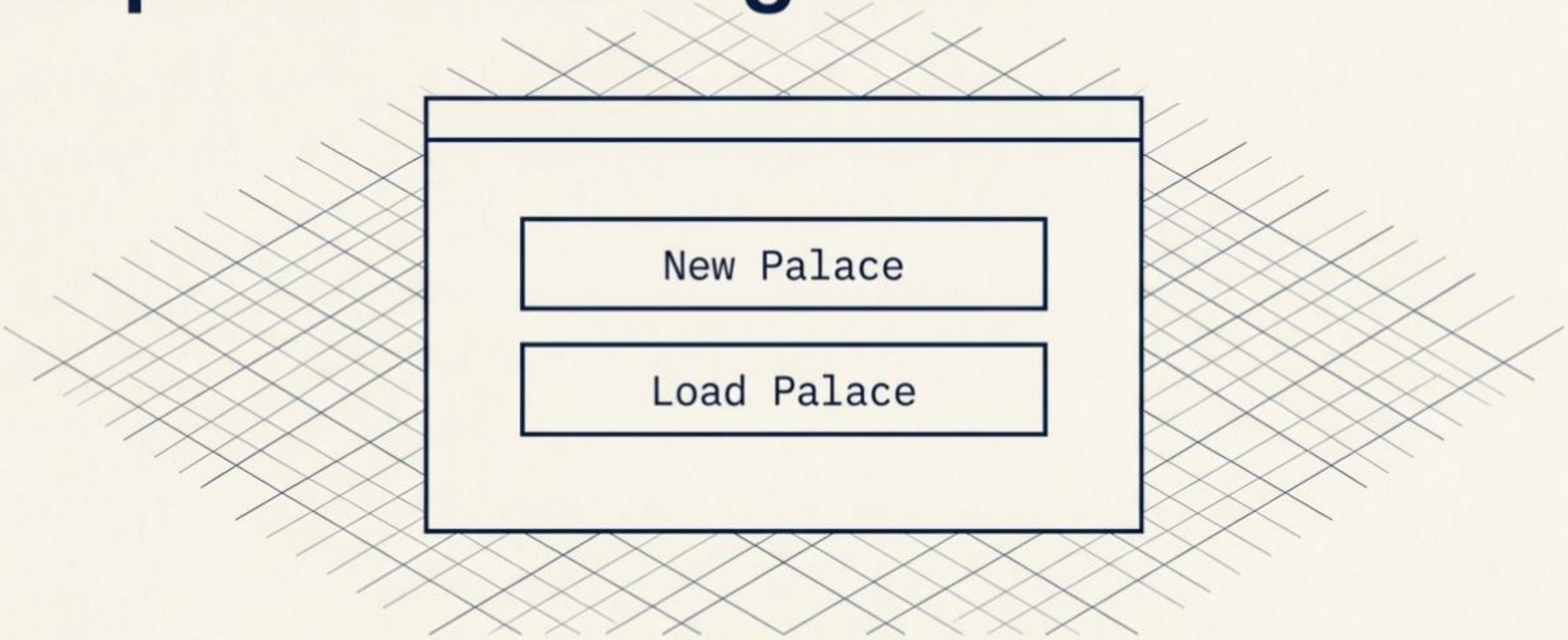
Reinforce: Recall is driven by spatial layout, color, labels, icons, and landmarks.

Spatial Architecture & Orientation



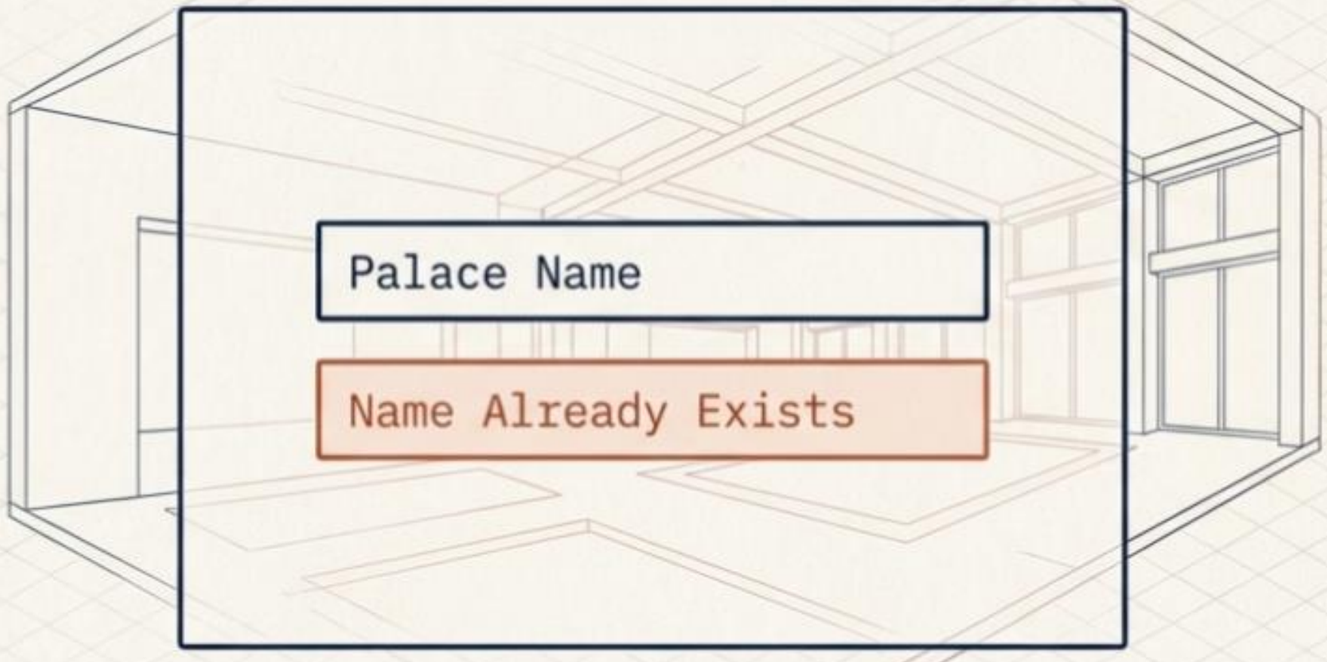
- **Central Entry Hall:** The primary palace organizer.
- **Unique Identity:** Each palace features a distinct central landmark.
- **Branching Rooms:** Creates logical separation between different topics.
- **Clear Wayfinding:** Every room has a unique name and visible entrance marker.

Step 1: The Starting Point



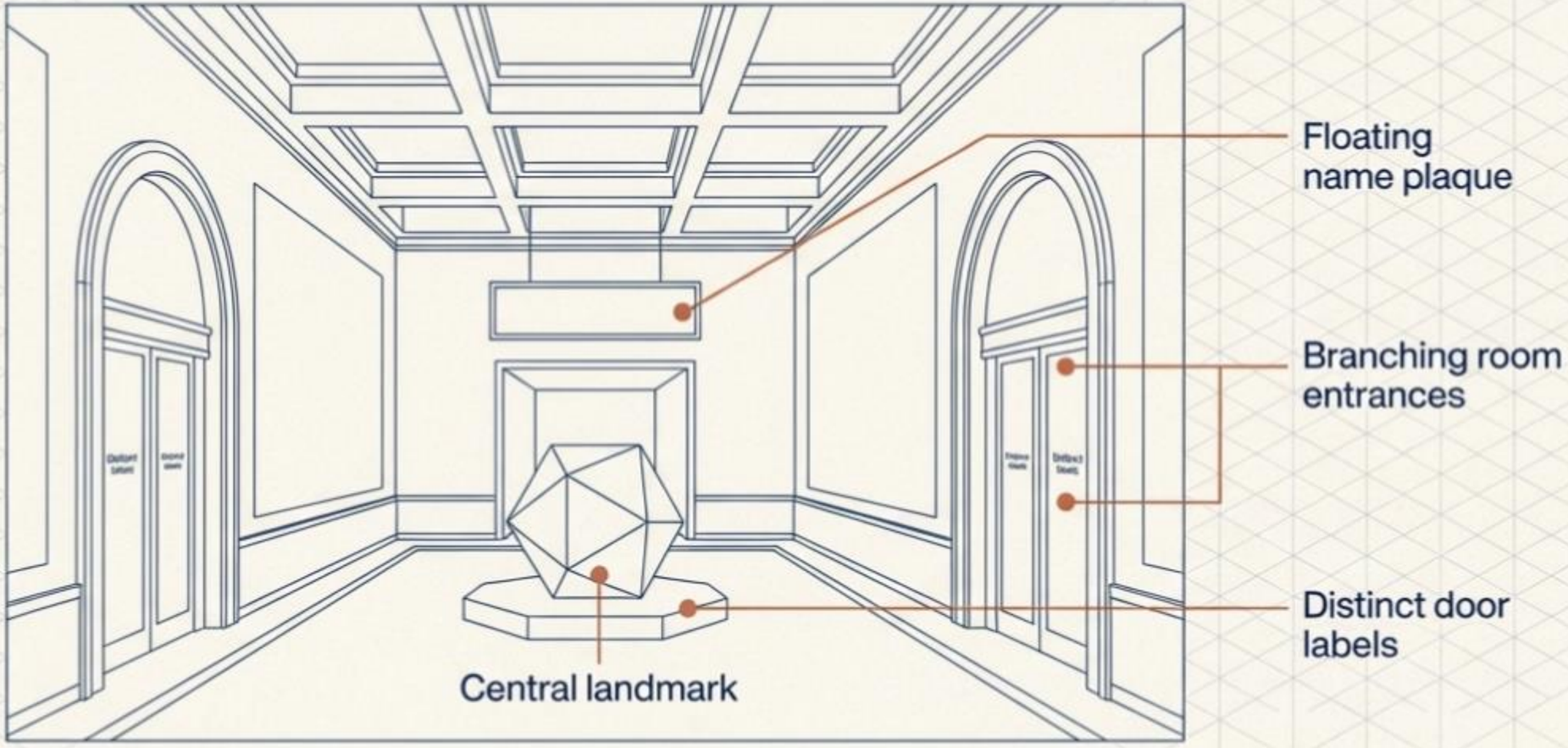
Project launches into the Main Menu. The first decision: Start fresh or continue previous work.

Step 2: Creation & Validation



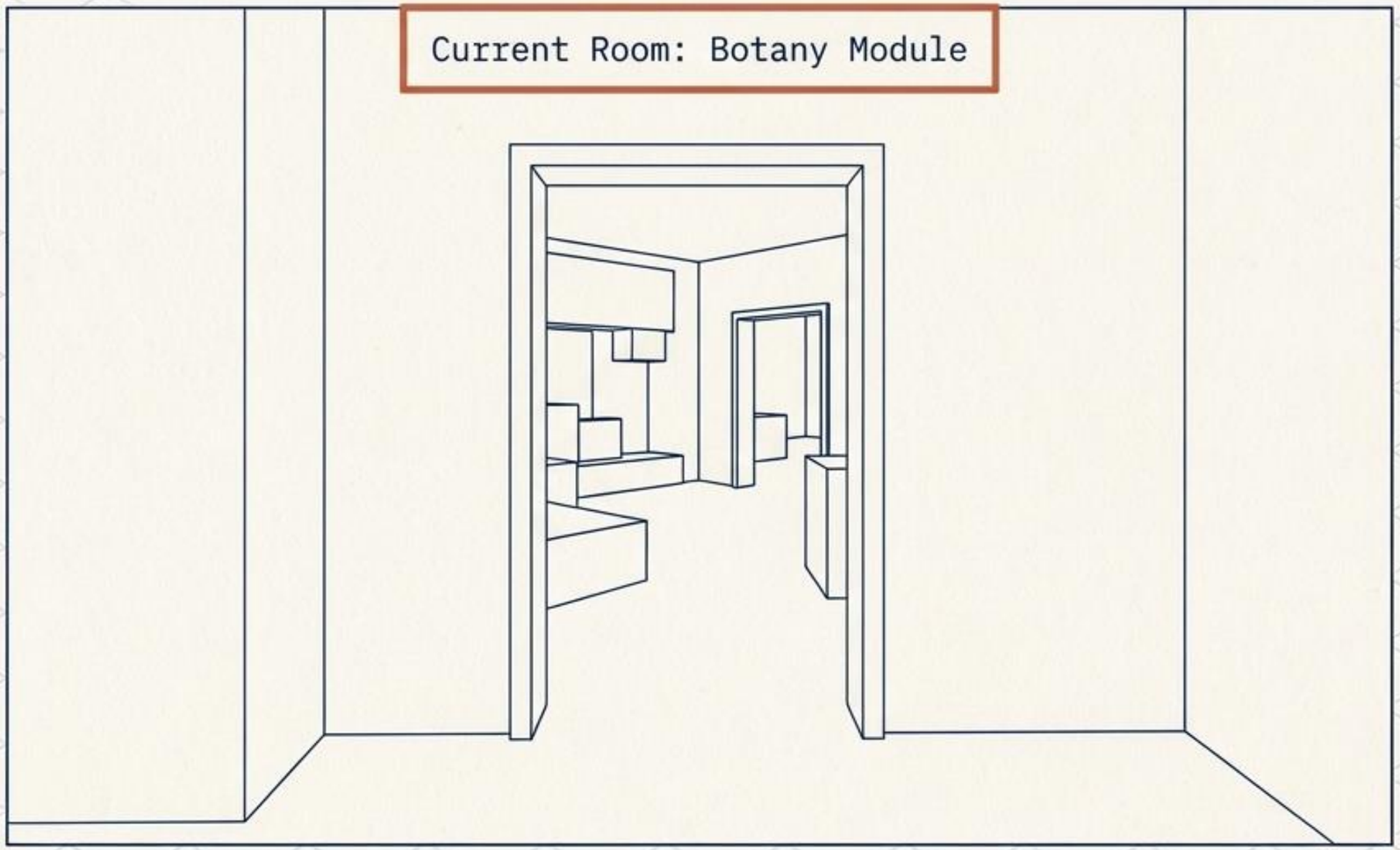
- User creates a new palace.
- System requires a unique name (prevents duplicates).
- Upon creation, the user spawns directly into the Entry Hall.

Step 3: Spatial Recognition



User immediately identifies their unique palace.

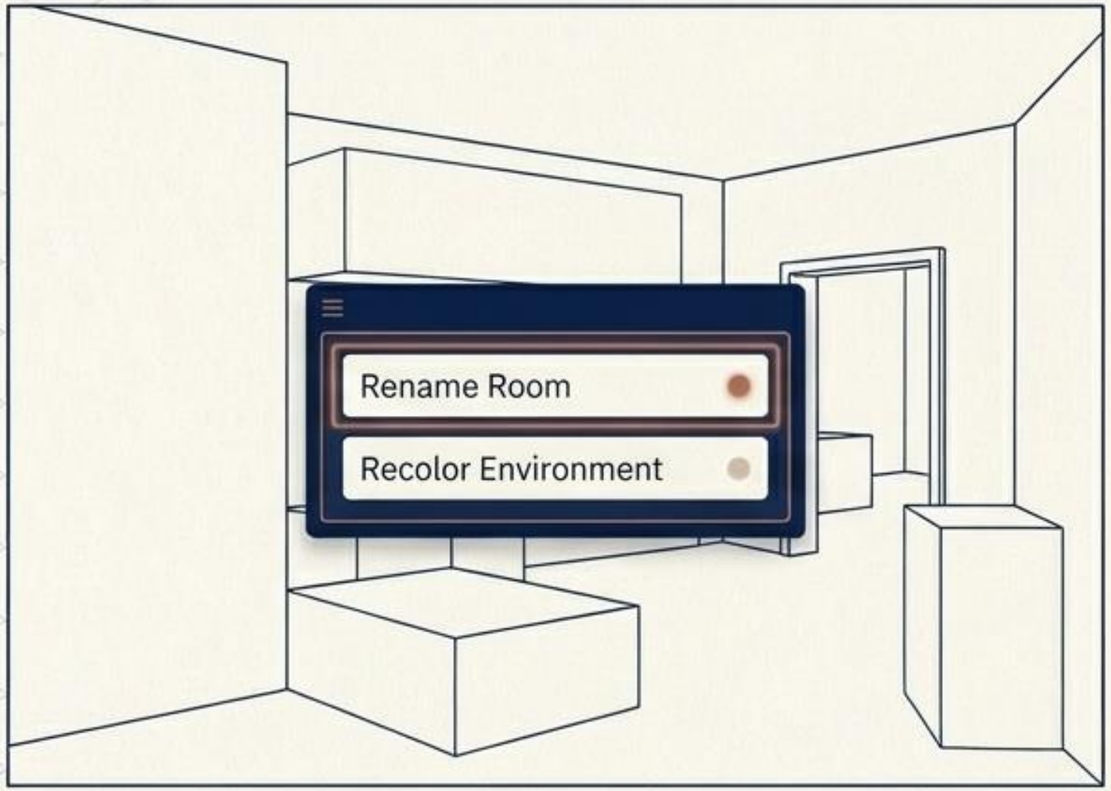
Step 4: Movement & Navigation



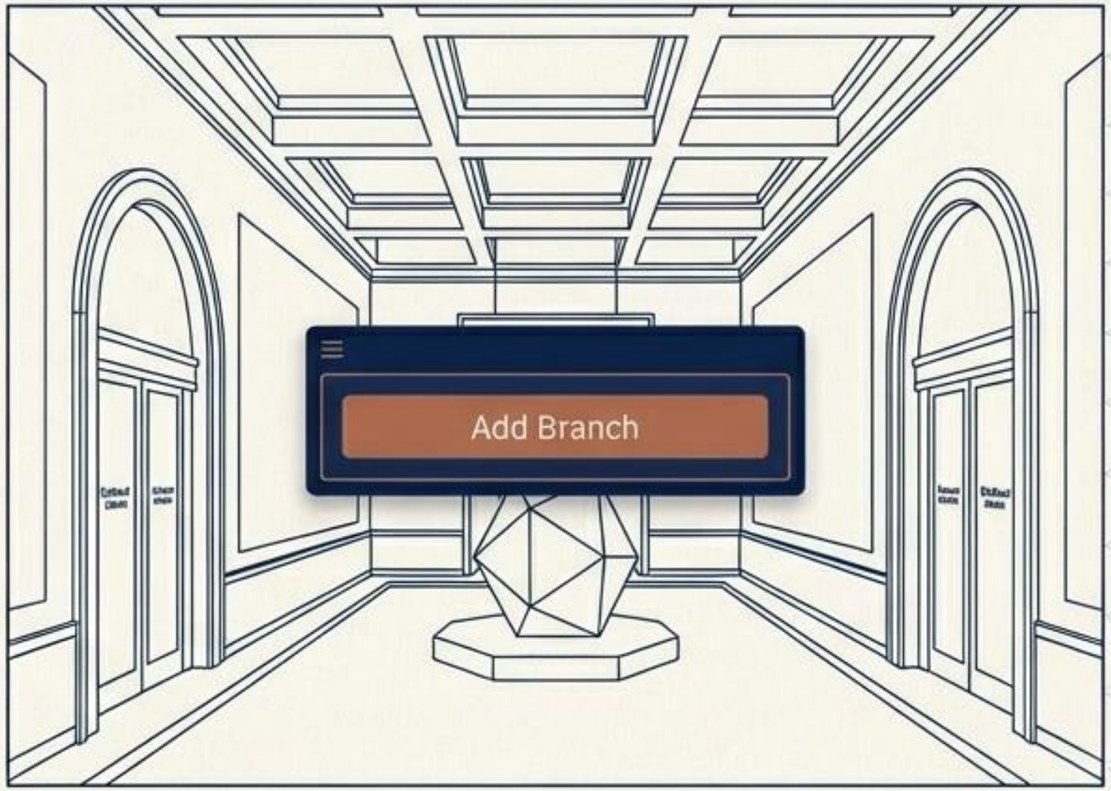
- User walks from the Entry Hall into a branch room.
- The HUD dynamically updates with the new room name.
- Navigation reinforces the hub-and-branch spatial structure.

Step 5: The Editing Loop

Press ‘E’ to edit the current space.



Inside Rooms: Rename and recolor.



In Entry Hall: Rename the entire palace and add one new room branch.

Step 6: Persistence

- Changes are saved implicitly during the editor flow.
- User safely returns to the Main Menu.
- Palace can be fully reloaded using JSON save data.

Milestone Deliverables Achieved

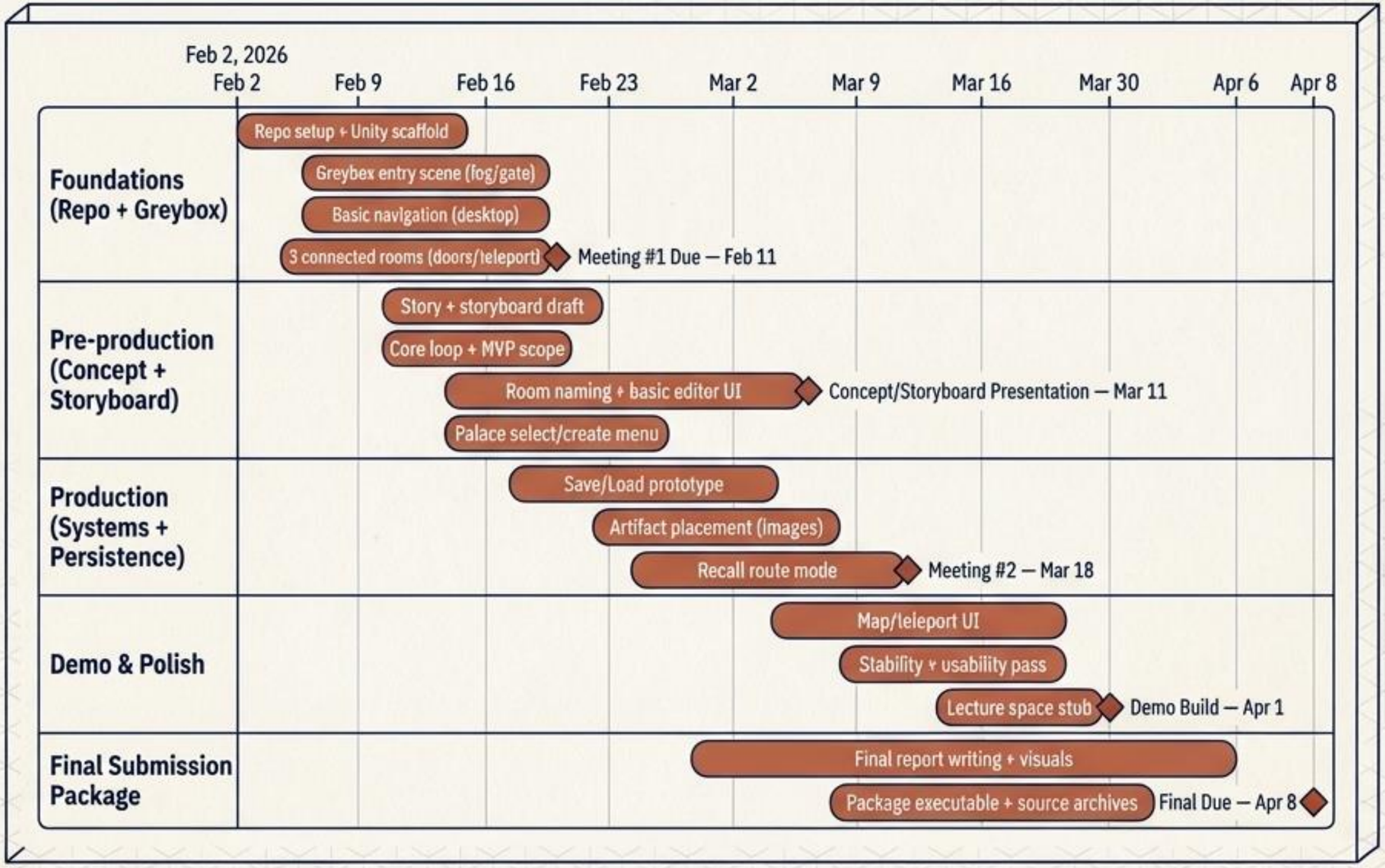
- ✔ First-person navigation & scene flow
- ✔ Named palace creation, loading, and deletion
- ✔ Duplicate-name prevention logic
- ✔ Room naming and environmental recoloring
- ✔ JSON data persistence

Project Roadmap: Winter 2026

Here is my updated Gantt plan.

This reflects my currently completed milestone.

As you can see, I have outlined what remains for my next review.

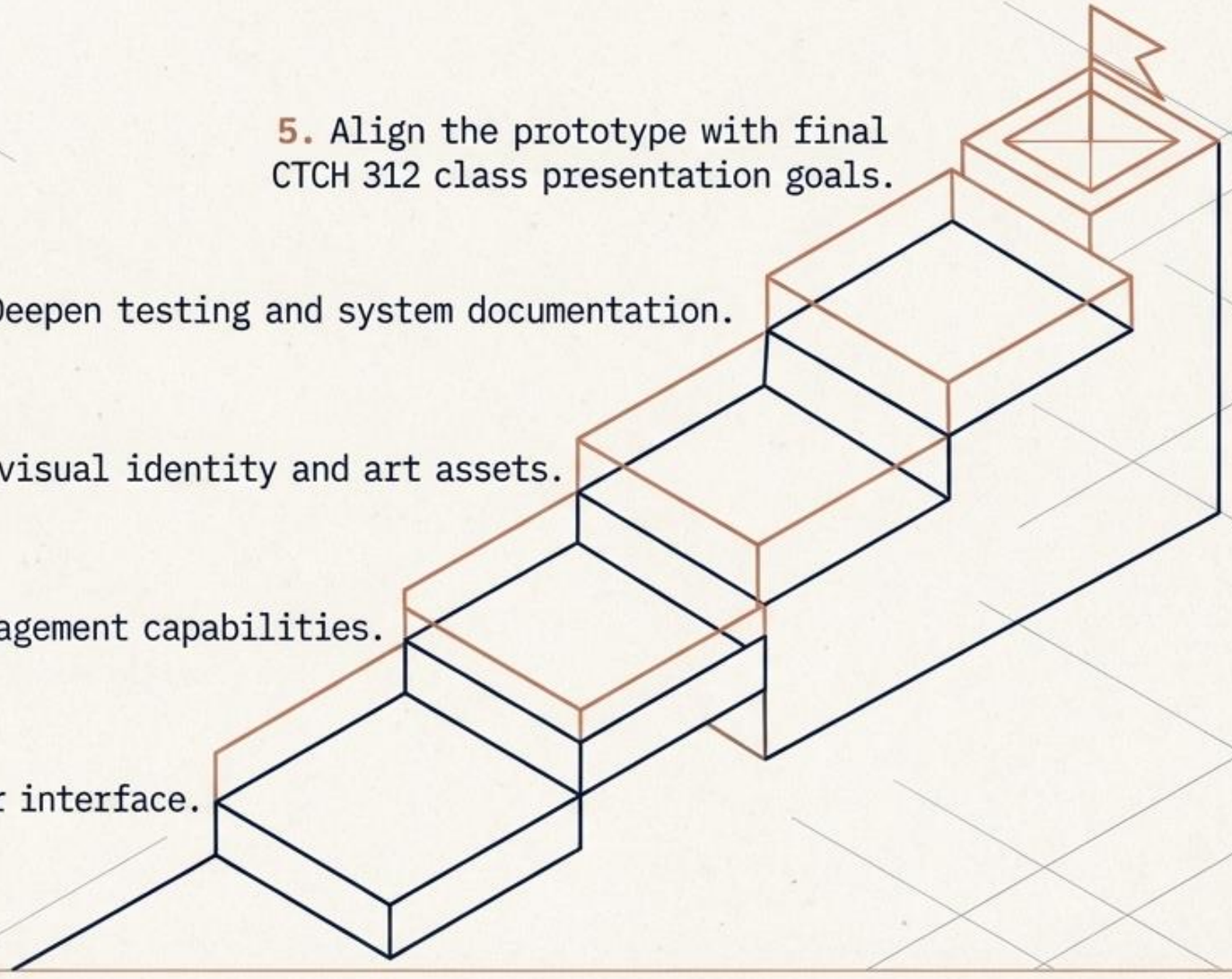


colored bars = tasks, diamonds = milestones

Current Scope & Constraints

Interface	Scope
Current UI is functional, but not final.	Editing is currently limited to adding one branch from the Entry Hall.
Aesthetics	**QA**
Art assets remain in the placeholder/greybox phase.	Testing and polish require further expansion.

The Path to Final Presentation



5. Align the prototype with final CTCH 312 class presentation goals.

4. Deepen testing and system documentation.

3. Improve visual identity and art assets.

2. Expand room management capabilities.

1. Polish the user interface.